CSA1607 - DWDM

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CO3 - AT2

Entropy formula:-

$$H(X) = -\sum_{i=1}^{m} P_i \log(P_i)$$

$$\text{Info}(D) = \sum_{i=1}^{m} P_i \log_2(P_i)$$

$$\text{Info}_A(D) = \sum_{j=1}^{v} \frac{|D_j|}{|D|} \times \text{Info}(D_j)$$

$$\text{Gain}(A) = \text{Info}(D) - \text{Info}(D_j)$$

→ class P :- buys_computer = "yes"
→ class N :- buys_computer = "No"

Age	class		I(P,N)
	P	N	
<30	2	3	0.9710
31-40	4	0	0
>40	3	2	0.9710
Total	9	5	0.9401

$$I(2,3) = -\sum P_i \log(P_i)$$

$$= -\frac{2}{5}\log_2\left(\frac{2}{5}\right) - \frac{3}{5}\log_2\left(\frac{3}{5}\right)$$

$$= -\frac{2}{5}(-1.322) - \frac{3}{5}(-0.737)$$

$$= 0.5288 + 0.4422$$

$$= 0.9710$$

$I(4,0) = -\frac{4}{4}(\log_2(1)) - 0$

$= 0$

$I(3,2) = -\frac{3}{5}\log_2(3/5) - \frac{2}{5}\log_2(2/5)$

$= -\frac{3}{5}(-0.737) - \frac{2}{5}(-1.322)$

$= 0.4422 + 0.5288$

$= 0.9710$

$I(9,5) = -\frac{9}{14}\log_2(9/14) - \frac{5}{14}\log_2(5/14)$

$= -9/14(-0.6374) - 5/14(-1.4854)$

$= 0.4097 + 0.5304$

$= 0.9401$

$Info_{age}(D) = \frac{5}{14} \times 0.971 + \frac{4}{14} \times 0 + \frac{5}{14} \times 0.971$

$= 0.3467 + 0 + 0.3467$

$= 0.6934$

$Gain(Age) = 0.9401 - 0.6934$

$= 0.2467$

Income	class		I(P,N)
	P	N	
high	2	2	1
medium	4	2	0.918
low	3	1	0.811
Total	9	5	0.9401

$I(2,2) = -\sum P_i \log_2(P_i)$

$= -\frac{2}{4}\log_2(2/4) - 2/4$

$= \log_2(2/4)$

$= -1/2(-1) - 1/2(-1)$

$= 1$

$I(4,2) = -4$

$I(4,2) = -\frac{4}{6}\log_2(4/6) - \frac{2}{6}\log_2(2/6)$

$= -\frac{4}{6}(-0.585) - \frac{2}{6}(-1.585)$

$= 0.39 + 0.528$

$= 0.918$

$I(3,1) = -\frac{3}{4}\log_2(3/4) - \frac{1}{4}\log_2(1/4)$

$= -3/4(-0.415) - 1/4(-2)$

$= 0.311 + 0.5$

$= 0.811$

$I(9,5) = -\frac{9}{14}\log_2(9/14) - 5/14\log_2(5/14)$

$= -9/14(-0.6324) - 5/14(-1.4854)$

$= 0.9401$

$Info_{age}(D) = \frac{4}{14}\times 1 + \frac{6}{14}\times 0.918 + \frac{4}{14}\times 0.811$

$= 0.285 + 0.393 + 0.231$

$= 0.909$

Gain(Age) $= 0.9401 - 0.909$

$= 0.0311$

Gain(student) = 0.151

Gain(credit = rating) = 0.048

Gain(income) = 0.0311

	gain	ratio	index
age	0.246	0.156	0.443
income	0.03	1.019	
student	0.151		
credit	0.48		

gain ratio $(A) = \text{Gain}(A) / \text{splitinfo}(A)$

$$\text{splitinfo}_A(D) = \sum_{j=1}^{n} \frac{|D_j|}{|D|} \times \log_2\left(\frac{|D_j|}{|D|}\right)$$

$$\text{Splitinfo}_{age} = -\frac{5}{14}\log_2\frac{5}{14} - \frac{4}{14}\log_2\frac{4}{14} - \frac{5}{14}\log_2\frac{5}{14}$$

$= 0.5304 + 0.5304 + 0.5163$

$= 1.5771 \approx 1.58$

$$\text{G ratio} = \frac{0.246}{1.58} = 0.156$$

$$\text{splitinfo}_{(income)} = -\frac{4}{14}\log_2\frac{4}{14} - \frac{6}{14}\log_2\frac{6}{14} - \frac{4}{14}\log_2\frac{4}{14}$$

$= 0.5163 + 0.5163 + 0.5238$

$= 1.5564$

gain ratio $= 1.019$

Gini index :-

$$\text{Gini}(D) = 1 - \sum_{j=1}^{n} p_j^2$$

$$\text{Gini}_A(D) = \frac{|D_1|}{|D|}\text{gini}(D_1) + \frac{|D_2|}{|D|}\text{gini}(D_2)$$

$$\text{Gini}(D) = 1 - \left[\left(\frac{9}{14}\right)^2 + \left(\frac{5}{14}\right)^2\right]$$

$= 1 - [0.413 + 0.127]$

$= 1 - 0.514$

$= 0.460$

partition the data set w.r.t income

$D_1 = \{low, medium\}$, $D_2 = \{High\}$

10 4

$$\text{Gini}_A(D) = \frac{10}{14}\left[1 - \left(\frac{7}{10}\right)^2 - \left(\frac{3}{10}\right)^2\right] + \frac{4}{14}\left[1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2\right]$$

$= 0.714(1 - 0.49 - 0.09) + 0.285[1 - 2.25 - 2.25]$

$= 0.714(0.42) + 0.285(1 - 0.5)$

$= 0.31 + 0.1425$

$= 0.44$

take low high together & medium seperate

$D_1 = \{low, high\}$ $D_2 = \{$ ~~Hig~~ medium $\}$

8   6

$$Gini_A(D) = \frac{8}{14}\left[1-\left(\frac{5}{8}\right)^2-\left(\frac{3}{8}\right)^2\right] + \frac{6}{14}\left[1-\left(\frac{4}{6}\right)^2-\left(\frac{2}{6}\right)^2\right]$$

$$= 0.571[1-0.39-0.14] + 0.428[1-0.43-0.10]$$

$$= 0.571(0.47) + 0.428(0.47)$$

$$= 0.257 + 0.2011$$

$$= 0.4581$$

take high medium together & low seperate

~~$Gini_A$~~ $(D_1) = \{high, medium\}$ $D_2 = \{low\}$

4

$$Gini_A(D) = \frac{10}{14}\left(1-\left(\frac{6}{10}\right)^2-\left(\frac{4}{10}\right)^2\right] + \frac{4}{14}\left[1-\left(\frac{3}{4}\right)^2-\left(\frac{1}{4}\right)^2\right]$$

$$= \frac{10}{14}(0.48) + \frac{4}{14}(0)$$

$$= 0.450$$

LM	H	0.443
LH	M	0.458
MH	L	0.450

confusion matrix :-

Actual class / predicted class	cancer = yes	cancer = NO	Total
cancer = yes	90(TP)	210(FN)	300(P)
cancer = no	140(FP)	9560(TN)	9700(N)
Total	230(P')	9770(N')	10000(all)

→ accuracy $= \frac{90+9560}{10000} = \frac{9650}{10000} \times 100 = 96.5$

→ Error rate = 100 − 96.5 (or) = 1 − 0.965

= 3.5 = 0.035 × 100

= 3.5

→ ~~(~~ sensitivity = $\frac{90}{30\cancel{0}}$ = 0.3 × 100 = 30

→ speificity = $\frac{956\cancel{0}}{970\cancel{0}}$ = 0.985 × 100 = 98.5

→ precision = $\frac{90}{90+210}$ = $\frac{9\cancel{0}}{30\cancel{0}}$ = 0.3 × 100 = 30

→ F = $\frac{2 \times 39.1 \times 30}{39.1 + 30}$ = $\frac{2346}{69.1}$ = 33.95%

1) A shepherd boy gets bored thending the town's flock. To have some fun he cries out, "wolf!" To have some is insight have some fun he cries out, protect the flock, but then get really the villagers run to a joke on them mad when they realise boy was playing

predicted

actual

TP wolf = yes	FN wolf = NO
FP wolf = NO	TN wolf = yes

2) Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images identity TP, FP, TN and calculate the precision & recall & F1 score

	predicted cat	predicted nocat	
actual cat	25	5	30
Actual nocat	20	50	70
	45	65	100

$\rightarrow$ precision :- $\dfrac{TP}{TP+FP} = \dfrac{25}{25+20} = \dfrac{25}{45} = 0.556$

$\rightarrow$ Recall $= \dfrac{TP}{TP+FN} = \dfrac{25}{25+5} = \dfrac{25}{30} = 0.8333$

$= 83.33\%$

$\rightarrow$ F_1 score $= \dfrac{2\,(\text{precision})(\text{Recall})}{\text{precision} + \text{Recall}}$

$= \dfrac{2(0.556)(0.8333)}{0.556 + 0.8333}$

$= 0.667$

F_1 score $= 66.67\%$

(3) An email service provider wants to classify emails as spam or not spam using features :

- contains "offer" (yes / No)
- contains "free" (yes / No)
- email length (short / long)

sample Data set

offer	free	length	class
yes	yes	short	class
yes	No	long	Not spam
No	yes	short	spam
No	No	long	Not spam
yes	yes	long	spam
No	No	short	Not spam

A) 1) Total spam = 6

spam = 3 , Not spam = 3

$p(\text{spam}) = \frac{3}{6} = 0.5$

$p(\text{Not spam}) = \frac{3}{6} = 0.5$

2) conditional probabilities:

for spam:-

$p(\text{otter} = \text{yes} / \text{spam}) = \frac{2}{3}$

$p(\text{free} = \text{yes} / \text{spam}) = \frac{3}{3} = 1$

$p(\text{length} = \text{short} / \text{spam}) = 2/3$

for not spam :-

$p(\text{cattar} = \text{yes} / \text{Not spam}) = 1/3$

$p(\text{free} = \text{yes} / \text{Not spam}) = 0/3 = 0$

$p(\text{length} = \text{short} / \text{Not spam}) = 1/3$

3) classity

cotter = yes , free = No , length = short)

for spam:-

$p(\text{spam} / x) \propto p(\text{spam}) \times p(\text{yes} / \text{spam}) \times p(\text{No} / \text{spam}) \times p(\text{short} / \text{spam})$

$= 0.5 \times 2/3 \times 0 \times 2/3 = 0$

for not spam

$p(\text{Not spam} / x) \propto 0.5 \times 1/3 \times 1 \times 1/3$

prediction = Not spam

Mooarid 18/8/28